

## CARAVELAS, Brazil

WMO: 834970

Lat: 17.652S

Lon: 39.253W

Elev: 11

StdP: 101.19

Time Zone: -3.00 (W03)

Period: 83-94

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 8             | 16.2       | 17.2 | 14.8                          | 10.5 | 20.3 | 15.8 | 11.2 | 20.0 | 10.3                  | 23.5 | 9.4 | 24.4 | 0.4                   | 240  | 0.280 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 7.8                    | 31.8            | 25.2 | 31.2 | 25.1 | 30.9 | 25.0 | 26.2                | 30.0 | 25.8 | 29.4 | 25.6 | 29.1 | 5.1                  | 60   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.2                            | 20.3 | 27.4 | 24.9 | 20.0 | 27.2 | 24.6 | 19.7 | 26.9 | 81.3          | 30.1 | 79.8 | 29.9 | 78.5 | 29.2 | 30.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.1          | 8.9  | 7.9 | 14.4                       | 34.6 | 1.0                | 1.8 | 13.6                                               | 35.9 | 13.1       | 37.0 | 12.5       | 38.0 | 11.8       | 39.4 |    |    |
| (5)               |      |     | 13.3                       | 28.3 | 1.5                | 1.2 | 12.2                                               | 29.2 | 11.4       | 29.9 | 10.5       | 30.6 | 9.5        | 31.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 24.8   | 26.6  | 26.9  | 26.8  | 26.0  | 24.6  | 23.1  | 22.4  | 22.3  | 23.2  | 24.3  | 25.3  | 25.9 |
|                                                                               | DBStd       | 2.06   | 0.89  | 0.81  | 0.91  | 1.05  | 1.29  | 1.26  | 1.27  | 1.32  | 1.43  | 1.46  | 1.44  | 1.24 |
|                                                                               | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDD10.0     | 5388   | 516   | 474   | 521   | 478   | 453   | 392   | 384   | 380   | 395   | 442   | 460   | 493  |
|                                                                               | CDD18.3     | 2346   | 257   | 241   | 262   | 228   | 194   | 142   | 126   | 122   | 145   | 184   | 210   | 235  |
|                                                                               | CDH23.3     | 17167  | 2381  | 2347  | 2483  | 1845  | 1249  | 674   | 506   | 510   | 721   | 1096  | 1497  | 1857 |
|                                                                               | CDH26.7     | 4418   | 739   | 823   | 840   | 552   | 287   | 92    | 46    | 33    | 73    | 173   | 306   | 454  |
| Wind                                                                          | WSAvg       | 2.7    | 3.0   | 2.4   | 2.6   | 2.2   | 2.1   | 1.8   | 2.0   | 2.4   | 3.2   | 3.4   | 3.8   | 3.5  |
| Precipitation                                                                 | PrecAvg     | 1430   | 102   | 79    | 136   | 139   | 126   | 87    | 115   | 70    | 83    | 139   | 196   | 149  |
|                                                                               | PrecMax     | 2291   | 398   | 320   | 535   | 312   | 334   | 246   | 288   | 203   | 267   | 404   | 464   | 404  |
|                                                                               | PrecMin     | 736    | 2     | 0     | 8     | 9     | 0     | 4     | 17    | 9     | 5     | 6     | 10    | 12   |
|                                                                               | PrecStd     | 333    | 94    | 71    | 104   | 74    | 72    | 50    | 66    | 44    | 59    | 95    | 111   | 86   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 32.1  | 32.2  | 32.2  | 31.9  | 31.0  | 29.2  | 28.6  | 28.4  | 28.9  | 30.2  | 30.6  | 31.2 |
|                                                                               |             | MCWB   | 25.4  | 25.4  | 25.1  | 25.1  | 25.3  | 23.8  | 23.2  | 22.9  | 23.3  | 24.2  | 25.1  | 25.2 |
|                                                                               | 2%          | DB     | 31.1  | 31.6  | 31.7  | 31.1  | 30.0  | 28.4  | 27.9  | 27.5  | 28.1  | 29.2  | 29.9  | 30.3 |
|                                                                               |             | MCWB   | 25.1  | 25.3  | 25.0  | 25.1  | 24.8  | 23.5  | 22.8  | 22.5  | 23.1  | 23.9  | 24.5  | 24.8 |
|                                                                               | 5%          | DB     | 30.6  | 31.1  | 31.1  | 30.5  | 29.2  | 27.9  | 27.1  | 26.9  | 27.6  | 28.2  | 29.1  | 29.9 |
|                                                                               |             | MCWB   | 24.9  | 25.1  | 25.0  | 25.0  | 24.5  | 23.3  | 22.6  | 22.2  | 22.8  | 23.5  | 24.2  | 24.7 |
|                                                                               | 10%         | DB     | 30.0  | 30.7  | 30.6  | 29.9  | 28.4  | 26.9  | 26.2  | 26.1  | 26.9  | 27.8  | 28.5  | 29.1 |
|                                                                               |             | MCWB   | 24.7  | 25.0  | 25.0  | 24.9  | 24.1  | 22.9  | 22.3  | 21.8  | 22.5  | 23.3  | 24.0  | 24.5 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 26.6  | 26.5  | 26.6  | 26.3  | 25.9  | 24.7  | 24.2  | 24.0  | 24.4  | 25.1  | 25.9  | 26.2 |
|                                                                               |             | MCDB   | 30.3  | 30.8  | 30.5  | 30.0  | 29.8  | 27.9  | 27.4  | 26.8  | 27.2  | 28.6  | 29.9  | 30.3 |
|                                                                               | 2%          | WB     | 25.9  | 26.0  | 26.0  | 25.7  | 25.3  | 24.1  | 23.5  | 23.2  | 23.8  | 24.5  | 25.2  | 25.5 |
|                                                                               |             | MCDB   | 29.5  | 29.9  | 29.6  | 29.3  | 28.7  | 27.2  | 26.6  | 26.2  | 26.6  | 27.8  | 28.4  | 28.9 |
|                                                                               | 5%          | WB     | 25.5  | 25.7  | 25.7  | 25.4  | 24.9  | 23.7  | 23.0  | 22.8  | 23.4  | 24.1  | 24.7  | 25.1 |
|                                                                               |             | MCDB   | 28.9  | 29.4  | 29.1  | 29.0  | 28.1  | 26.6  | 26.0  | 25.7  | 26.1  | 27.2  | 27.7  | 28.2 |
|                                                                               | 10%         | WB     | 25.2  | 25.3  | 25.4  | 25.1  | 24.5  | 23.3  | 22.6  | 22.3  | 23.1  | 23.7  | 24.5  | 24.7 |
|                                                                               |             | MCDB   | 28.4  | 29.0  | 28.8  | 28.6  | 27.5  | 26.0  | 25.3  | 25.0  | 25.6  | 26.6  | 27.3  | 27.6 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 7.1   | 7.8   | 7.7   | 7.6   | 7.7   | 7.9   | 7.8   | 7.7   | 7.1   | 6.3   | 6.0   | 6.3  |
|                                                                               |             | MCDBR  | 7.8   | 8.4   | 8.6   | 8.7   | 8.7   | 9.1   | 9.1   | 8.9   | 7.8   | 7.2   | 7.1   | 7.4  |
|                                                                               |             | MCWBR  | 2.6   | 2.7   | 2.8   | 3.3   | 3.8   | 4.4   | 4.5   | 4.2   | 3.3   | 2.9   | 2.7   | 2.7  |
|                                                                               | 5% WB       | MCDBR  | 7.2   | 7.8   | 7.6   | 8.1   | 8.1   | 8.2   | 8.4   | 8.3   | 7.0   | 6.4   | 6.3   | 6.7  |
|                                                                               |             | MCWBR  | 2.8   | 3.0   | 2.9   | 3.5   | 3.8   | 4.4   | 4.4   | 4.3   | 3.3   | 3.0   | 2.7   | 2.7  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.395  | 0.388 | 0.387 | 0.382 | 0.376 | 0.363 | 0.360 | 0.375 | 0.431 | 0.434 | 0.419 | 0.415 |      |
|                                                                               | taud        | 2.502  | 2.525 | 2.546 | 2.548 | 2.545 | 2.569 | 2.560 | 2.490 | 2.331 | 2.354 | 2.431 | 2.444 |      |
|                                                                               | Ebn at Noon | 950    | 947   | 925   | 893   | 862   | 857   | 870   | 886   | 870   | 893   | 921   | 931   |      |
|                                                                               | Edh at Noon | 115    | 112   | 106   | 99    | 93    | 87    | 90    | 103   | 129   | 131   | 123   | 122   |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 6.61   | 6.45  | 5.77  | 4.96  | 4.17  | 3.88  | 4.12  | 4.79  | 5.17  | 5.61  | 5.57  | 6.17  |      |
|                                                                               | RadStd      | 0.57   | 0.42  | 0.44  | 0.34  | 0.21  | 0.24  | 0.22  | 0.23  | 0.43  | 0.74  | 0.65  | 0.61  |      |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                       |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page